



**School of Electronics Engineering**  
**CONTINUOUS ASSESSMENT TEST - I**  
**WINTER SEMESTER 2024-2025**

**Programme Name & Branch** : B.Tech ECE  
**Course Code and Course Name** : BECE204L-Microprocessors and Microcontrollers  
**Faculty Name(s)** : Dr.T.N.PADMINI (Course Coordinator)  
**Class Number(s)** : VL2024250502421/2423/2424/2425/2426  
**Date of Examination** : 30/1/2025  
**Exam Duration** : 90 minutes **Maximum Marks: 50**

**General instruction(s):**

- Answer All Questions
- M - Max mark; CO – Course Outcome; BL – Blooms Taxonomy Level (1 – Remember, 2 – Understand, 3 – Apply, 4 – Analyse, 5 – Evaluate, 6 – Create)
- Course Outcomes  
CO3 - Comprehend the architectures and programming of 8051 microcontroller  
CO4 - Deploy the implementation of various peripherals such as general purpose input/output, timers, serial communication, LCD, keypad and ADC with 8051 microcontroller

| Q. No        | Question                                                                                                                                                                                                                                                                                                                                                                                                      | M  | CO            | BL          |   |           |   |     |   |     |   |     |   |              |   |     |   |   |   |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|---------------|-------------|---|-----------|---|-----|---|-----|---|-----|---|--------------|---|-----|---|---|---|
| 1.           | <b>a) Identify, if the following 8051 instructions have any error. If so, give the correct instructions:</b><br>MOV A,P0.1<br>DEC DPTR<br>INC #40H<br>SETB A<br>CLRB P2.0                                                                                                                                                                                                                                     | 5  | 3             | 2           |   |           |   |     |   |     |   |     |   |              |   |     |   |   |   |
|              | <b>b) Find the total execution time of the following program, if the crystal frequency is 15 MHz.</b> <table><tr><td></td><td>Machine Cycle</td></tr><tr><td>MOV R3,#50H</td><td>1</td></tr><tr><td>HERE: NOP</td><td>1</td></tr><tr><td>NOP</td><td>1</td></tr><tr><td>NOP</td><td>1</td></tr><tr><td>NOP</td><td>1</td></tr><tr><td>DJNZ R3,HERE</td><td>2</td></tr><tr><td>RET</td><td>2</td></tr></table> |    | Machine Cycle | MOV R3,#50H | 1 | HERE: NOP | 1 | NOP | 1 | NOP | 1 | NOP | 1 | DJNZ R3,HERE | 2 | RET | 2 | 5 | 3 |
|              | Machine Cycle                                                                                                                                                                                                                                                                                                                                                                                                 |    |               |             |   |           |   |     |   |     |   |     |   |              |   |     |   |   |   |
| MOV R3,#50H  | 1                                                                                                                                                                                                                                                                                                                                                                                                             |    |               |             |   |           |   |     |   |     |   |     |   |              |   |     |   |   |   |
| HERE: NOP    | 1                                                                                                                                                                                                                                                                                                                                                                                                             |    |               |             |   |           |   |     |   |     |   |     |   |              |   |     |   |   |   |
| NOP          | 1                                                                                                                                                                                                                                                                                                                                                                                                             |    |               |             |   |           |   |     |   |     |   |     |   |              |   |     |   |   |   |
| NOP          | 1                                                                                                                                                                                                                                                                                                                                                                                                             |    |               |             |   |           |   |     |   |     |   |     |   |              |   |     |   |   |   |
| NOP          | 1                                                                                                                                                                                                                                                                                                                                                                                                             |    |               |             |   |           |   |     |   |     |   |     |   |              |   |     |   |   |   |
| DJNZ R3,HERE | 2                                                                                                                                                                                                                                                                                                                                                                                                             |    |               |             |   |           |   |     |   |     |   |     |   |              |   |     |   |   |   |
| RET          | 2                                                                                                                                                                                                                                                                                                                                                                                                             |    |               |             |   |           |   |     |   |     |   |     |   |              |   |     |   |   |   |
| 2.           | Develop an 8051-assembly language program to get a byte of data from port P1. If the Most Significant Bit (MSB) of the data is 1, then generate a square wave of 25% duty cycle on P2.1. Otherwise, generate a waveform of 60% duty cycle on P2.1.                                                                                                                                                            | 10 | 4             | 3           |   |           |   |     |   |     |   |     |   |              |   |     |   |   |   |
| 3.           | Write an assembly program to add 10 bytes data using loop instructions and store the lower byte of the result in R2 and higher byte in R3. The bytes are stored in ROM space starting at 300H as shown below:<br>ORG 300H<br>DB 92H,34H,84H,29H,22H,11H,33H,44H,55H,66H                                                                                                                                       | 10 | 3             | 3           |   |           |   |     |   |     |   |     |   |              |   |     |   |   |   |



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| 4.      | <p><b>Show the data in various memory locations and stack pointer after executing each and every instruction of the following 8051 code</b></p> <table><tr><th>Line No</th><th>Program</th><th>Data in RAM(H)</th></tr><tr><td>1.</td><td>MOV 05H,#3EH</td><td>05H :     ; SP :</td></tr><tr><td>2.</td><td>MOV 0AH,#DFH</td><td>0AH :     ; SP :</td></tr><tr><td>3.</td><td>MOV 2FH,#45H</td><td>2FH :     ; SP :</td></tr><tr><td>4.</td><td>MOV SP, #69H</td><td>SP :</td></tr><tr><td>5.</td><td>PUSH 05H</td><td>05H :     ; SP :</td></tr><tr><td>6.</td><td>PUSH 0AH</td><td>0AH :     ; SP :</td></tr><tr><td>7.</td><td>PUSH 2FH</td><td>2FH :     ; SP :</td></tr><tr><td>8.</td><td>POP 4AH</td><td>4AH :     ; SP :</td></tr><tr><td>9.</td><td>POP 4BH</td><td>4BH :     ; SP :</td></tr><tr><td>10.</td><td>PUSH 4BH</td><td>4BH :     ; SP :</td></tr><tr><td>11.</td><td>POP 5AH</td><td>5AH :     ; SP :</td></tr><tr><td></td><td>END</td><td></td></tr></table> | Line No          | Program | Data in RAM(H) | 1. | MOV 05H,#3EH | 05H :     ; SP : | 2. | MOV 0AH,#DFH | 0AH :     ; SP : | 3. | MOV 2FH,#45H | 2FH :     ; SP : | 4. | MOV SP, #69H | SP : | 5. | PUSH 05H | 05H :     ; SP : | 6. | PUSH 0AH | 0AH :     ; SP : | 7. | PUSH 2FH | 2FH :     ; SP : | 8. | POP 4AH | 4AH :     ; SP : | 9. | POP 4BH | 4BH :     ; SP : | 10. | PUSH 4BH | 4BH :     ; SP : | 11. | POP 5AH | 5AH :     ; SP : |  | END |  | 10 | 3 | 3 |
|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|---------|----------------|----|--------------|------------------|----|--------------|------------------|----|--------------|------------------|----|--------------|------|----|----------|------------------|----|----------|------------------|----|----------|------------------|----|---------|------------------|----|---------|------------------|-----|----------|------------------|-----|---------|------------------|--|-----|--|----|---|---|
| Line No | Program                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Data in RAM(H)   |         |                |    |              |                  |    |              |                  |    |              |                  |    |              |      |    |          |                  |    |          |                  |    |          |                  |    |         |                  |    |         |                  |     |          |                  |     |         |                  |  |     |  |    |   |   |
| 1.      | MOV 05H,#3EH                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 05H :     ; SP : |         |                |    |              |                  |    |              |                  |    |              |                  |    |              |      |    |          |                  |    |          |                  |    |          |                  |    |         |                  |    |         |                  |     |          |                  |     |         |                  |  |     |  |    |   |   |
| 2.      | MOV 0AH,#DFH                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 0AH :     ; SP : |         |                |    |              |                  |    |              |                  |    |              |                  |    |              |      |    |          |                  |    |          |                  |    |          |                  |    |         |                  |    |         |                  |     |          |                  |     |         |                  |  |     |  |    |   |   |
| 3.      | MOV 2FH,#45H                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 2FH :     ; SP : |         |                |    |              |                  |    |              |                  |    |              |                  |    |              |      |    |          |                  |    |          |                  |    |          |                  |    |         |                  |    |         |                  |     |          |                  |     |         |                  |  |     |  |    |   |   |
| 4.      | MOV SP, #69H                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | SP :             |         |                |    |              |                  |    |              |                  |    |              |                  |    |              |      |    |          |                  |    |          |                  |    |          |                  |    |         |                  |    |         |                  |     |          |                  |     |         |                  |  |     |  |    |   |   |
| 5.      | PUSH 05H                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 05H :     ; SP : |         |                |    |              |                  |    |              |                  |    |              |                  |    |              |      |    |          |                  |    |          |                  |    |          |                  |    |         |                  |    |         |                  |     |          |                  |     |         |                  |  |     |  |    |   |   |
| 6.      | PUSH 0AH                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 0AH :     ; SP : |         |                |    |              |                  |    |              |                  |    |              |                  |    |              |      |    |          |                  |    |          |                  |    |          |                  |    |         |                  |    |         |                  |     |          |                  |     |         |                  |  |     |  |    |   |   |
| 7.      | PUSH 2FH                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 2FH :     ; SP : |         |                |    |              |                  |    |              |                  |    |              |                  |    |              |      |    |          |                  |    |          |                  |    |          |                  |    |         |                  |    |         |                  |     |          |                  |     |         |                  |  |     |  |    |   |   |
| 8.      | POP 4AH                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 4AH :     ; SP : |         |                |    |              |                  |    |              |                  |    |              |                  |    |              |      |    |          |                  |    |          |                  |    |          |                  |    |         |                  |    |         |                  |     |          |                  |     |         |                  |  |     |  |    |   |   |
| 9.      | POP 4BH                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 4BH :     ; SP : |         |                |    |              |                  |    |              |                  |    |              |                  |    |              |      |    |          |                  |    |          |                  |    |          |                  |    |         |                  |    |         |                  |     |          |                  |     |         |                  |  |     |  |    |   |   |
| 10.     | PUSH 4BH                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 4BH :     ; SP : |         |                |    |              |                  |    |              |                  |    |              |                  |    |              |      |    |          |                  |    |          |                  |    |          |                  |    |         |                  |    |         |                  |     |          |                  |     |         |                  |  |     |  |    |   |   |
| 11.     | POP 5AH                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 5AH :     ; SP : |         |                |    |              |                  |    |              |                  |    |              |                  |    |              |      |    |          |                  |    |          |                  |    |          |                  |    |         |                  |    |         |                  |     |          |                  |     |         |                  |  |     |  |    |   |   |
|         | END                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                  |         |                |    |              |                  |    |              |                  |    |              |                  |    |              |      |    |          |                  |    |          |                  |    |          |                  |    |         |                  |    |         |                  |     |          |                  |     |         |                  |  |     |  |    |   |   |
| 5.      | <p><b>(i)Analyse the following 8051 code and determine the content of the register A, memory location 20H and carry flag after the execution of the following code.</b></p> <p>ORG 000H<br/>MOV A,#83H<br/>RLC A<br/>RLC A<br/>MOV 20H, #87H<br/>ANL C,02H<br/>END</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 5                |         |                |    |              |                  |    |              |                  |    |              |                  |    |              |      |    |          |                  |    |          |                  |    |          |                  |    |         |                  |    |         |                  |     |          |                  |     |         |                  |  |     |  |    |   |   |
|         | <p><b>(ii) Analyse the following 8051 code and determine the content of the registers A, R0, memory location 40H and carry flag after the execution of the following code.</b></p> <p>ORG 000H<br/>MOV A,#80H<br/>MOV R0,#40H<br/>MOV @R0,A<br/>INC A<br/>RRC A<br/>ADD A,40H<br/>END</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 5                | 3       | 2              |    |              |                  |    |              |                  |    |              |                  |    |              |      |    |          |                  |    |          |                  |    |          |                  |    |         |                  |    |         |                  |     |          |                  |     |         |                  |  |     |  |    |   |   |

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